

# Curriculum Vitae

Shu Hu

PhD Student · Department of Microbiology · Fudan University  
shu25@m.fudan.edu.cn | <https://hushuzuiniu.github.io/>

## EDUCATIONAL BACKGROUND

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|-----------------------------------------------------------------------------------|--------------------------------|
| <b>Fudan University</b>                                                           | Sep 2025 – Jun 2029 (expected) |
| PhD student in Microbiology   Department of Microbiology, School of Life Sciences |                                |
| <b>The University of Edinburgh</b>                                                | Sep 2021 – Nov 2022            |
| MSc in Bioinformatics, with Merit                                                 |                                |
| <b>Sichuan Agricultural University</b>                                            | Sep 2016 – Jun 2020            |
| BSc in Biological Science   GPA: 3.8/5.0; average mark: 88/100.                   |                                |
| English: CET-6 535; IELTS 6.0.                                                    |                                |

**RESEARCH FOCUS:** AI foundation models · Sequence representation learning · Deep learning · Viral evolution

## PUBLICATIONS

- **Hu S**, Ye D, Liao M, Xu T, Fan T, Dong N, Cui J. Paired Env-antibody representation learning predicts HIV-1 neutralization and traces antibody escape across viral evolution. *Cell Reports Methods*. 2026. **Under revision**
- Ye D, Hu F, Hu H, **Hu S**, Tan Y, Ouyang W, Li SZ, Cui J, Dong N. ViroBench: Benchmarking Nucleotide Foundation Models on Viral Genomics Tasks. *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. 2026; 12679–12690. doi: [10.1145/3770855.3819057](https://doi.org/10.1145/3770855.3819057)
- Schwab LSU, Xie R, Reilly E, Aban M, **Hu S**, Spirason N, Deng YM, et al. Epistasis facilitates the long-term antigenic evolution of the influenza B virus hemagglutinin. *bioRxiv*. 2026. doi: [10.64898/2026.07.22.739994](https://doi.org/10.64898/2026.07.22.739994)
- Xie R, Zhukova A, Peña PG, Iglesias G, **Hu S**, Wang J, Tsang TK, Dhanasekaran V, Kraemer MUG, Pybus OG, Gascuel O. Scalable deep-learning-based inference of time-varying transmission dynamics from outbreak phylogenies. *medRxiv*. 2026. doi: [10.64898/2026.05.07.26352673](https://doi.org/10.64898/2026.05.07.26352673)
- Siegers JY, Xie R, Edwards KM, Byrne AMP, **Hu S**, et al. Resurgence of Zoonotic Highly Pathogenic Avian Influenza A(H5N1) Virus in Cambodia. *New England Journal of Medicine*. 2025; 393(16):1650–1652. doi: [10.1056/NEJMc2504302](https://doi.org/10.1056/NEJMc2504302)
- **Hu S**, Cui J. Cattle H5N1 outbreak in US driven by cow’s “cross-nursing” behaviour: elucidating a novel transmission mechanism. *Emerging Microbes & Infections*. 2025; 14(1):2547736. doi: [10.1080/22221751.2025.2547736](https://doi.org/10.1080/22221751.2025.2547736)
- Yang K, Nizami S, **Hu S**, Zou L, Deng H, Xie J, Guo Q, Edwards KM, Dhanasekaran V, Yen HL, Wu J. Genetic diversity of highly pathogenic avian influenza H5N6 and H5N8 viruses in poultry markets in Guangdong, China, 2020–2022. *Journal of Virology*. 2025; 99(1):e01145–24. doi: [10.1128/jvi.01145-24](https://doi.org/10.1128/jvi.01145-24)
- Xie R, Adam DC, **Hu S**, Cowling BJ, Gascuel O, Zhukova A, Dhanasekaran V. Integrating Contact Tracing Data to Enhance Outbreak Phylodynamic Inference: A Deep Learning Approach. *Molecular Biology and Evolution*. 2024; 41(11):msae232. doi: [10.1093/molbev/msae232](https://doi.org/10.1093/molbev/msae232)

## RESEARCH AND PROFESSIONAL EXPERIENCE

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### Fudan University

Sep 2025 – present

#### PhD Research

- Co-developed the UniCAV model and training framework; curated datasets, conducted computational analyses and benchmarks, and drafted the manuscript (Cell Reports Methods, under revision).
- Contributed to ViroBench, a benchmark study of nucleotide foundation models published at KDD 2026.

### The University of Hong Kong

Jun 2023 – Jun 2024

#### Research Assistant I

- Conducted bioinformatics analyses of influenza and RSV, integrating genomic, clinical and immunological datasets.
- Contributed to research integrating contact-tracing data with phylogenetic analysis for outbreak inference; co-authored the resulting Molecular Biology and Evolution paper (2024).
- Participated in genomic epidemiology studies of avian influenza in Cambodia and southern China, contributing to publications in NEJM and Journal of Virology (2025).

### Shanghai Rightongene Technology Co., Ltd.

2024 – May 2025; Sep 2022 – May 2023

#### Bioinformatics Engineer

- Developed analysis methods for BCL6, BCL2 and MYC gene fusions in DLBCL sequencing workflows, addressing discrepancies between NGS and FISH results.
- Established low-depth whole-genome sequencing workflows for copy-number analysis using QDNAseq, Control-FREEC and ichorCNA.
- Built upstream and downstream single-cell RNA-seq workflows and collaborated with IT colleagues to deploy analyses on an automated platform.
- Developed RNA-seq expression-based models for leukemia molecular subtyping.

### The University of Edinburgh

Feb – Sep 2022

#### MSc Research

- Developed reusable Shiny modules and R scripts for interactive exploration of RNA-seq datasets during MSc study at the University of Edinburgh.

### Sichuan WoWente Biotechnology Co., Ltd.

May 2020 – May 2021

#### Reagent Developer

- Contributed to sex-hormone immunoassay reagent development and an inhibin A (INH-A) project.

## TECHNICAL SKILLS

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**AI and foundation models:** Nucleotide foundation-model benchmarking; protein language models; biological sequence representations.

**Deep learning:** Model development, training, evaluation, benchmarking and interpretation.

**Programming and computing:** R, Python, Linux, Shell, Docker, SQL.

**Bioinformatics:** Phylogenetics, BEAST, Nextstrain, RNA-seq, single-cell RNA-seq, WGS and copy-number analysis.

## RESEARCH SOFTWARE

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[github.com/hushuzuiniu](https://github.com/hushuzuiniu)

UniCAV · Single-cell RNA-seq workflows · RNA-seq molecular subtyping · R Shiny tools